

Use-Case Flow Specification

Project: Municipal Infrastructure Damage Reporting App

Use Case ID: UC-01

Use Case Name: Report Infrastructure Damage

Primary Actor: Citizen Reporter

1. Preconditions

1. The Citizen Reporter has access to the reporting portal.
2. The Citizen Reporter has infrastructure damage to report.
3. The Citizen Reporter has a photograph showing the reported damage.

2. Postconditions

Success

- A damage report is successfully submitted.
- The uploaded damage photograph is attached to the report.
- GPS information from the photograph is used to identify the incident location.
- The report is dispatched to the relevant municipal engineering department.

Failure

- The report is not submitted if required information is missing or invalid.

3. Main Success Scenario

1. The **Citizen Reporter** opens the infrastructure damage reporting function.
2. The system displays the damage report form.
3. The Citizen Reporter enters the required information about the infrastructure damage.
4. The Citizen Reporter uploads a photograph showing the damage.
5. The system validates the uploaded photograph.
6. The system extracts GPS information from the photograph.
7. The system identifies the incident location on the municipal GIS map.
8. The Citizen Reporter submits the damage report.

9. The system automatically dispatches the report to the relevant municipal engineering department.
10. The system confirms that the damage report has been successfully submitted.

4. Alternate Flow

A1 — Invalid or Unsupported Photo

1. At **Step 5**, the system determines that the uploaded photograph is invalid or unsupported.
2. The system does not attach the photograph to the report.
3. The system informs the Citizen Reporter that a valid photograph is required.
4. The Citizen Reporter uploads a valid photograph.
5. The flow resumes at **Step 5** of the Main Success Scenario.

5. Use-Case Relationships

UC-01 includes:

- **UC-02 — Upload Damage Photo**
- **UC-03 — Extract GPS Location**
- **UC-04 — Dispatch Damage Report**